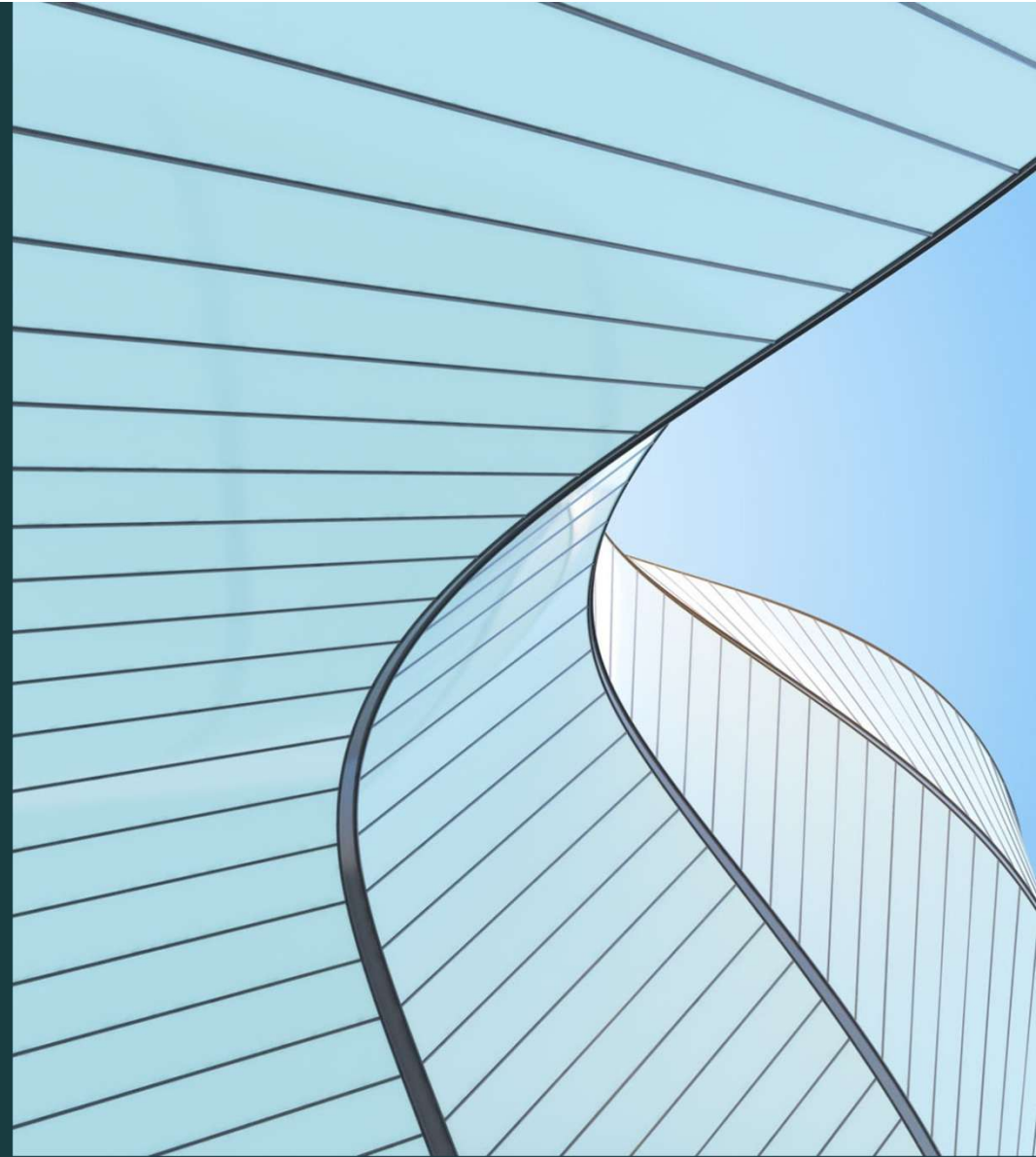


Team Caterpillar
10/15

ARINC 429 Midterm Presentation

This presentation outlines a the concept security gateway design of ARINC 429 data bus.



Our *team*



Joshua



Nate



Brandon

Content *overview* ARINC 429

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| Replay | DoS | Conclusion |



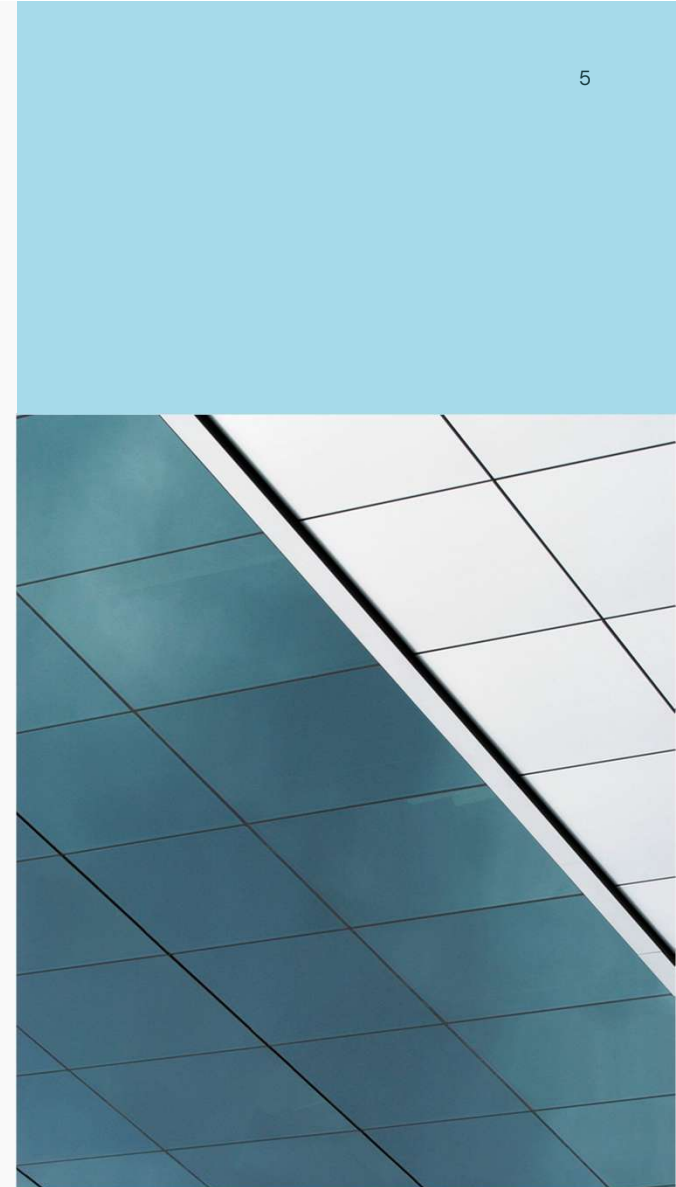
What is ARINC 429?

The ARINC 429 is a data bus used in commercial, private, and military aircrafts to transport avionic data throughout the plane.

This includes all of the flight data like the radar, relative pitch and yaw, airspeed, data to the black box and much more information.

How Does ARINC work?

- 1 transmitter sends data to up to 20 receivers through a single twisted wire pair in which it is unidirectional.
- The LRUs transmits 32-bit words, in which 24 of those bits contain data and 8 bits describe the data.
- There are no addresses for the LRUs but equipment IDs
- Transmission rates can either be 12.5 kHz or 100 kHz with a deviation of 1 and only one of these rates can be used at a time.



How Does ARINC work Continued

32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
P	SSM	Data																		LSB	SDI	Label									

ARINC 429 BNR Word Format

8	7	6	5	4	3	2	1	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
Label								SDI	LSB	Data																MSB				SSM	P

ARINC 429 Word Transfer Order

P (parity): P bit ensures there are an odd number of 1's. If there are an even number of ones that means the word was damaged in transit.

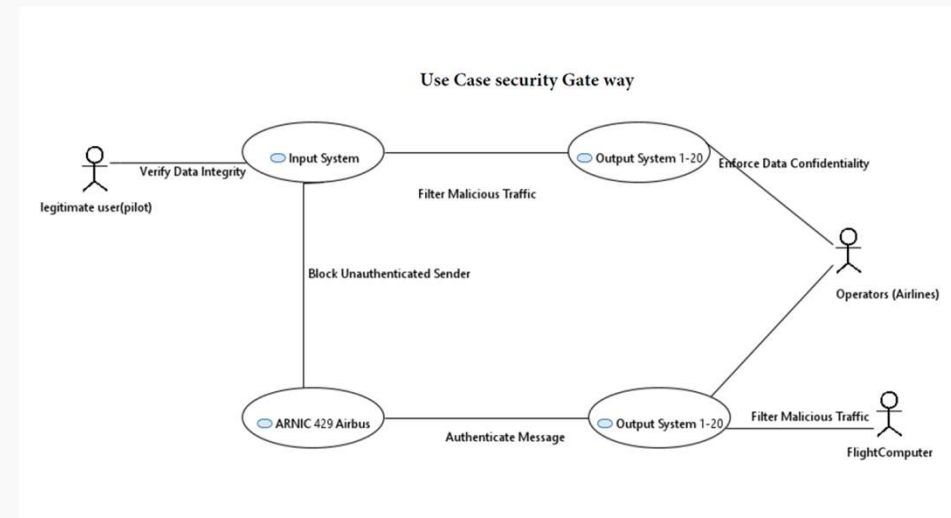
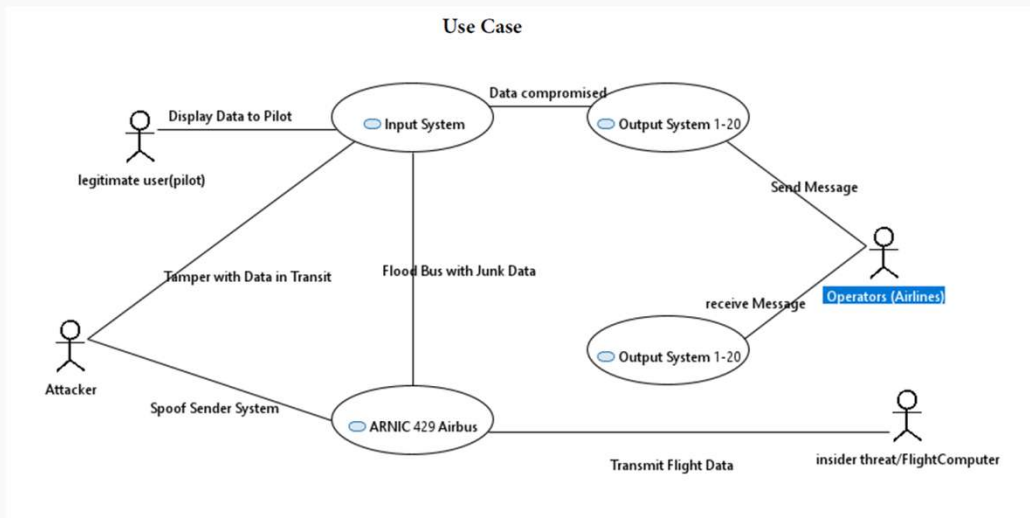
SSM: Provides context for the data (cardinal direction, positive or negative, ect)

Data: The data that the receiving LRU interprets

SDI: Informs the LRU of all receivers

Label: Informs LRU on what type of data (velocity, radar, intercom, ect)

SYSMML model (Use case)



Normal Operations:

- Transmit flight data
- Display data to the Pilot
- Identified Attack: spoof, Flood Bus...

Security Gateway Mitigations:

- Authenticate to verify sender identity.
- Verify data integrity to check for tampering.
- Filter Malicious Traffic to stop bus flooding.
- Enforce Data confidentiality.

Threat Modeling

	Denial of Service (DoS)	Altered Frames (Tampering)	Replay (Spoofing)
Damage	(3) Could cause systems to revert to reversion mode	(3) Can affect all receiving nodes on the same bus	(3) Can affect all receiving nodes on the same bus
Reproducibility	(2) The attacker needs access to the plane in order to carry on with the DoS attack but once they have it the it is easily repeatable	(1) Frame specifications varies by manufacturer	(3) does not require knowledge of data specifications, attack stays the same if the same LRU is compromised each time
Exploitability	(2) Needs physical access or privileged access to the avionics	(1) Requires physical access, custom malware, and knowledge of how receivers interpret data	(2) Requires physical access and custom malware
Affected users	(3) Potential for all passengers to be affected	(3) Potential for all passengers to be affected	(3) Potential for all passengers to be affected
Discoverability	(3) The protocol is public, but the bus architecture, label and data specifications are not public	(2) Knowledge of theoretical attack is easily discoverable due to public protocol information, however label and data specifications are not public	(2) Knowledge of theoretical attack is easily discoverable due to public protocol information, however label and data specifications are not public
Total	13(High)	10 (Medium)	13 (High)

Spoofing Attack

Attack Explanation

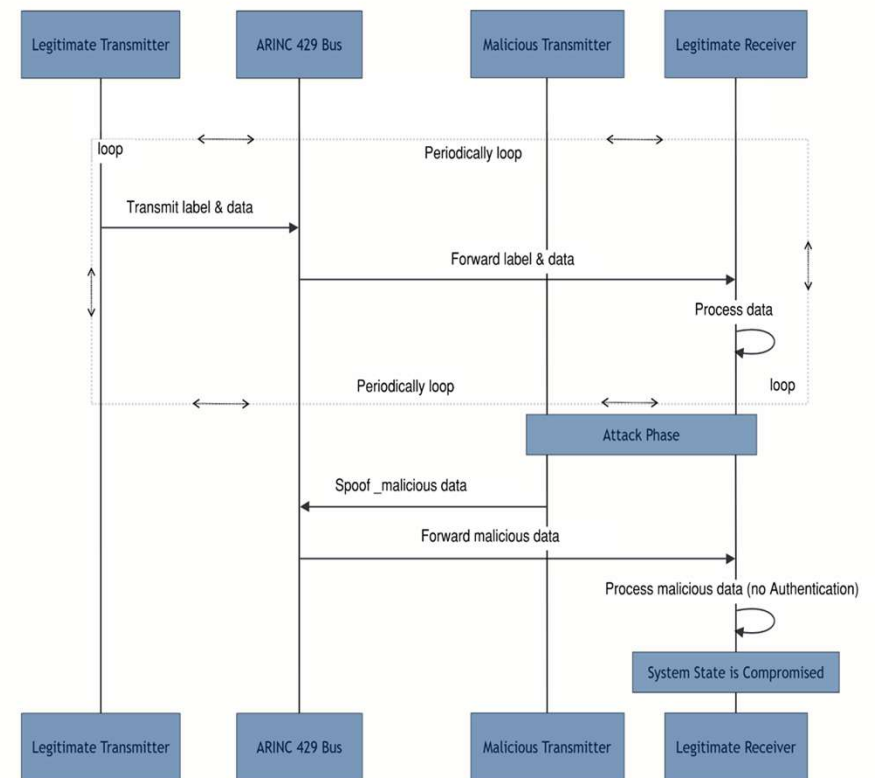
Diagram illustrates the normal flow of ARINC 429 communication.

A malicious transmitter then joins the bus and sends a message with an identical label.

Receiver can't distinguish the legitimate from the malicious.

Accepts and processes the spoofed data, leading to a compromised system.

SysML



Spoofing Defence

Defence Explanation

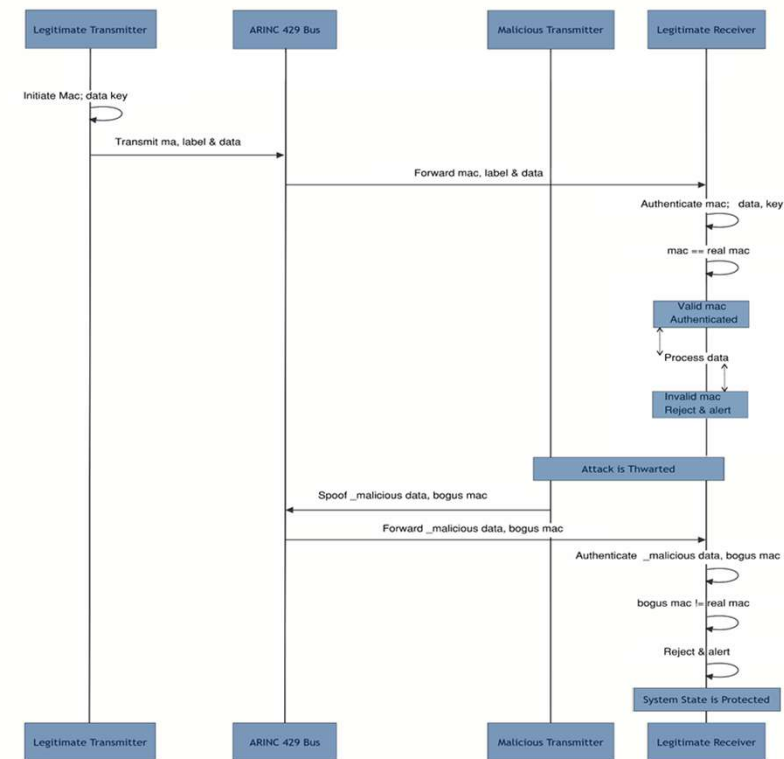
Diagram illustrates proposed cryptographic authentication defense prevents the spoofing attack.

legitimate transmitter now generates a MAC using the shared secret key and appends it to the message.

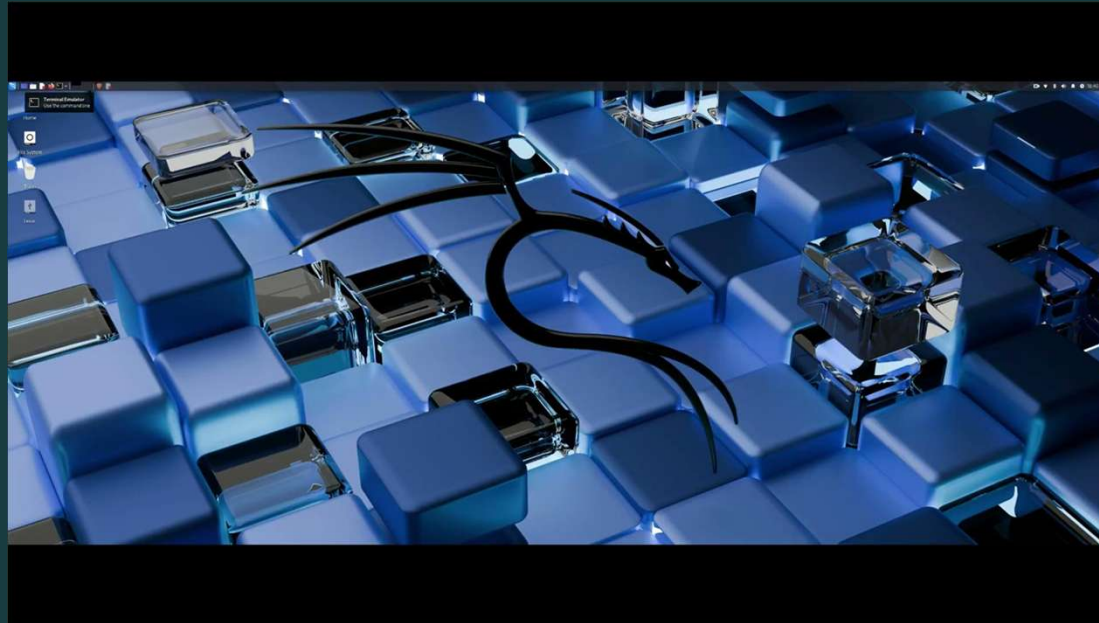
Receiver validates the MAC (fake or no MAC or real) before processing the data.

Accepts real data, malicious data rejected, and an alert can be triggered.

SysML



Spoofing Demo



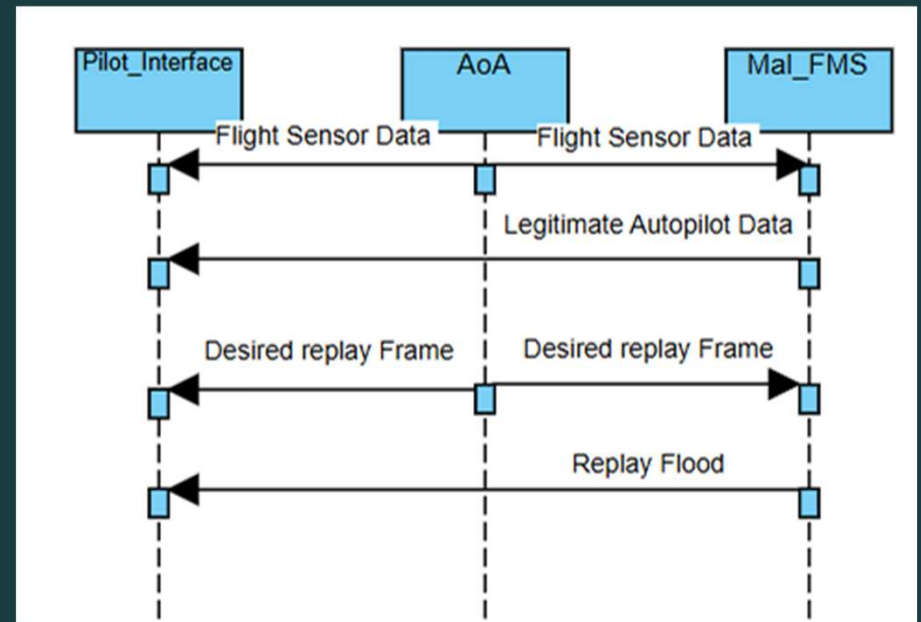
Replay Attack

Attack Explanation

The Attack works by having the malicious flight management capture its desired replay frame, and replay it over and over again to the pilot interface.

Because the Pilot interface has no authentication step, it accepts the malicious frames and allows for the attack to happen.

SysML



Replay Defence

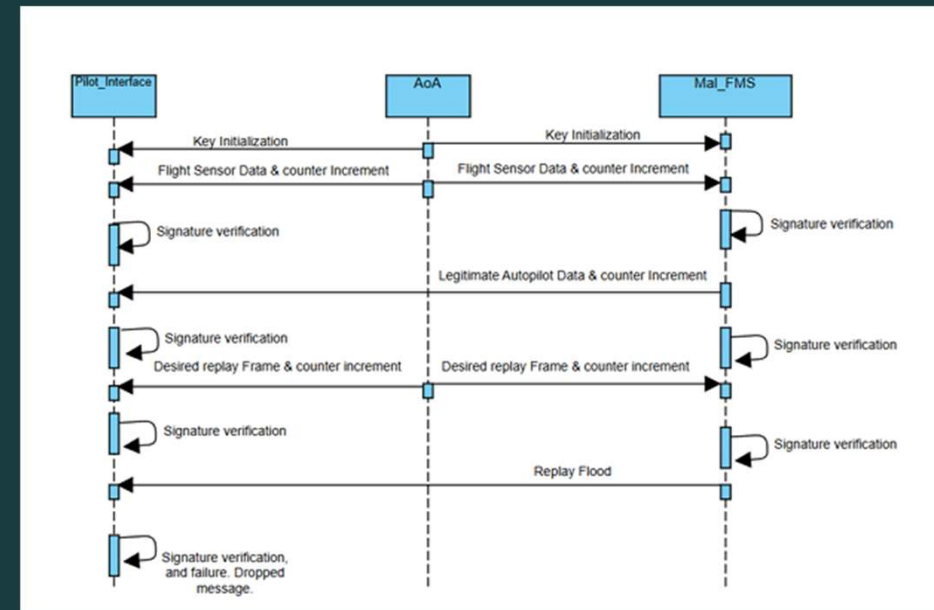
Defence Explanation

Similarly to the Spoofing defence a shared key is generated and to create a MAC that is validated by the receiver to validate the authentication of the user.

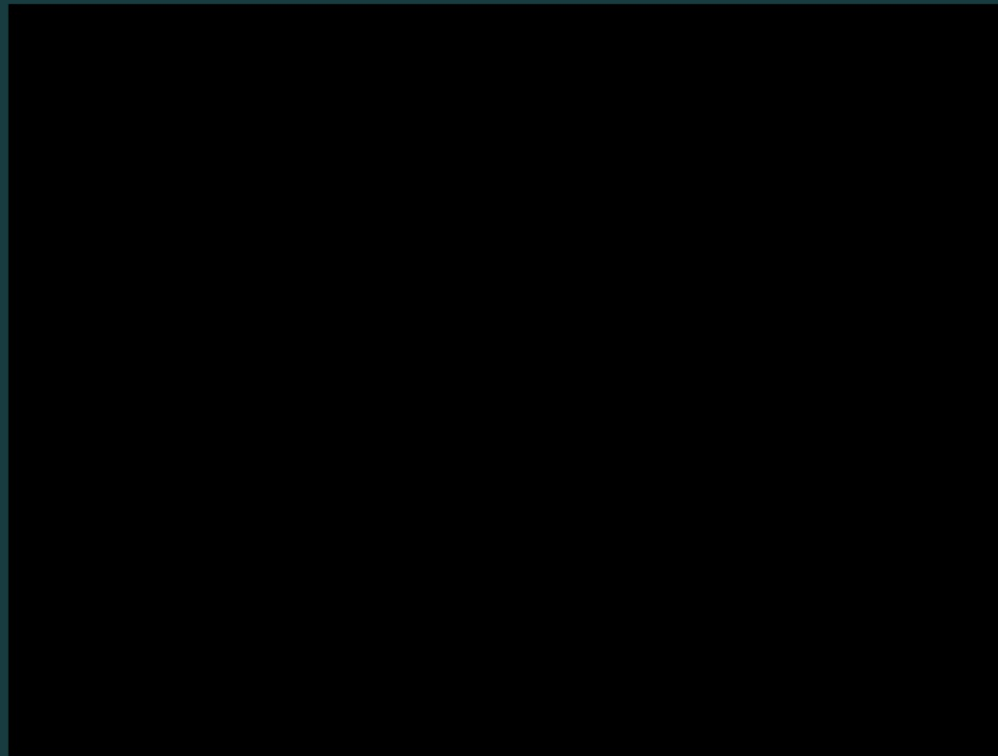
Additionally a NONCE is created through the use of a simple counter and is appended to a new field in the word (making a 40 bit word)

The receiving end checks the NONCE to ensure it is not a replay attack.

SysML



Replay Demo



DoS Attack

Attack Explanation

The diagram to the right shows roughly how a DoS attack would perform in an ARINC 429 databus

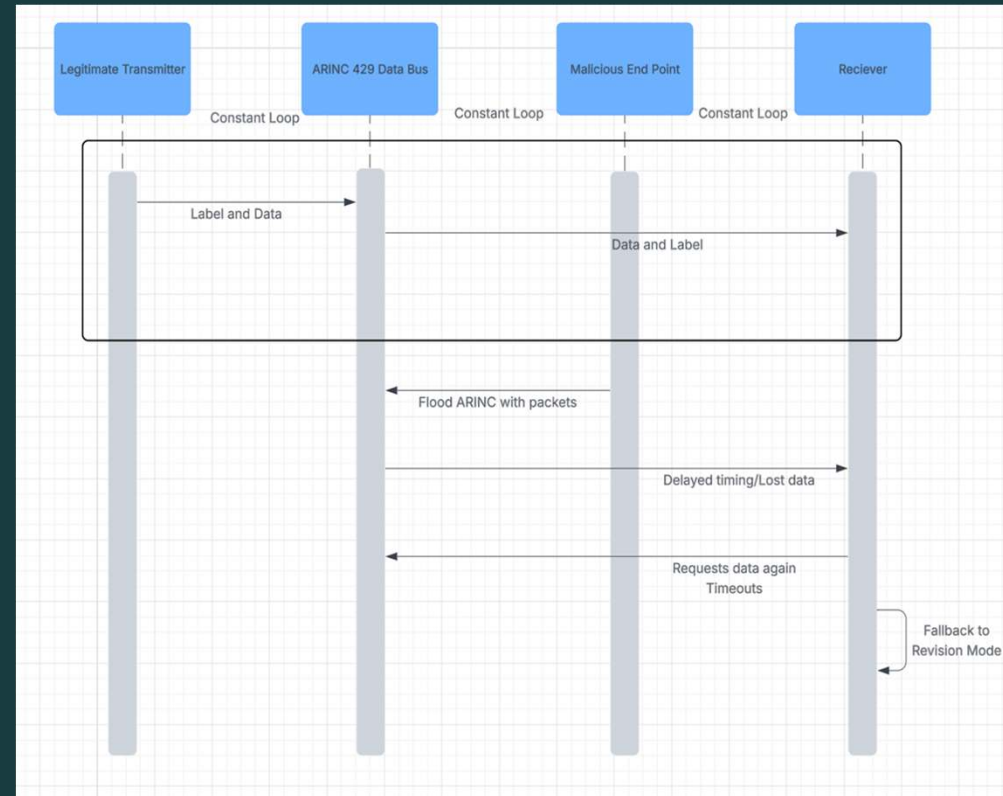
Firstly, the attacker would need to have physical access to the wire and have a node connecting them to the databus.

Next, the malicious end point would flood the data bus with packets to the point where they are unable to send data properly, which results in delayed timing and loss of data.

Then, the receiver will tell the databus that the data was corrupted or uncomplete, in which the receiver will request for the databus to transmit again

Finally, the receiver will get a timeout error and will then go into revision mode to try and protect itself.

SysML



DoS Defence

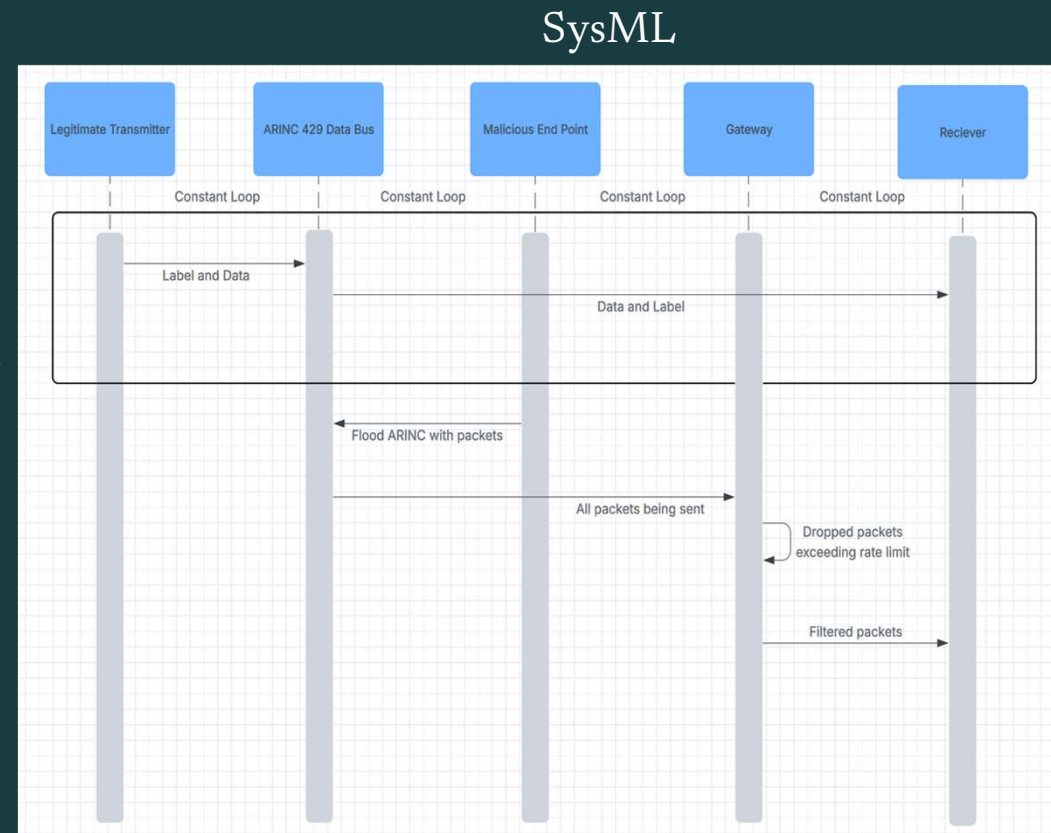
Defence Explanation

This defense differs to the ones before, where we are implementing a gateway between the data bus and the receiver, which is exemplified in the diagram to the right.

First the attacker floods the data bus with packets, and instead of the data bus sending them directly to the receiver, they go through a gateway.

Then, this gateway enforces rate limits per transmitter and creates a buffer of packets being sent ensuring timing is correct.

Finally, the filtered packets are then sent from the gateway to the receiver and the receiver then processes these packets.



Conclusion & Lessons Learned

Recommendations

- Add a MAC signature to verify authentication
- Add a NONCE to ensure there is no replay
- Actively Monitor, and ensures compliance security requirements.
- Add a gateway to drop packets that exceed the rate parameters

Challenges

- Flashing the key to each system
- Checking and monitoring everyone that have access to the bus.
- Identifying breaches during operations

References and Questions?

- [1] <https://github.com/JoshuaMFrench/CYSE465-Caterpillars-Group>
- [2] B. Waldmann. ARINC 429 Tutorial, v2.2. (2019). Accessed: September 12th, 2025. [Online]. Available: <https://www.aim-online.com/wp-content/uploads/2019/07/aim-tutorial-oview429-190712-u.pdf>
- [3] <https://www.omg.org/sysml/INCOSE-OMGSysML-Tutorial-Final>



Individual Contribution

Josh

- Initial case study for the proposal with team
- Replay attack
- Replay Defense
- Replay SysML
- Replay Demo
- Final Report

Nate

- Initial case study for the proposal with team.
- System Architecture (Sysml model).
- Spoofing attack
- Spoofing Defense
- Spoofing SysML
- Spoofing Demo
- Final report

Brandon

- Initial case study for the proposal with team
- DoS Attack
- Dos Defense
- Dos SysML
- Final Report

Team Effort

It was “All hand on deck” approach for the project in the beginning. Once we figured out our attack vectors, we then split into our our sections. Josh did the replay attack, Nate did the spoofing attack, and Brandon did the DoS attack. Once that was completed, we came together to write the report and presentation

Thank you